

Stable liquid phases and the lever rule

Read and prepare: 15–20 minutes total. Bring tentative answers; uncertainty is useful for discussion.

Why can an equal-fugacity solution still need a stability check?

Candidates and stable equilibrium

At fixed T and P , stable equilibrium minimizes total Gibbs energy over admissible states. Equality of chemical potentials identifies candidate coexisting phases. A trivial solution with identical compositions is not a split. A local minimum or a converged equation solver need not identify the globally stable state.

The common tangent

For a binary molar mixing-Gibbs curve $g(x)$, two coexisting compositions share a tangent. The tangent must support the entire admitted curve from below. The straight line represents the molar Gibbs energy attainable by mixing amounts of the two phases. Curvature concerns local stability; comparison with all admissible phase combinations concerns global stability.

Amounts follow composition

At given coexistence conditions, changing overall composition inside the tie line changes phase amounts while the endpoints stay fixed. This separates the equilibrium calculation from the material-balance calculation. A fit at one temperature does not establish temperature dependence. Replicate measurements of one tie line also do not create additional equilibrium states.

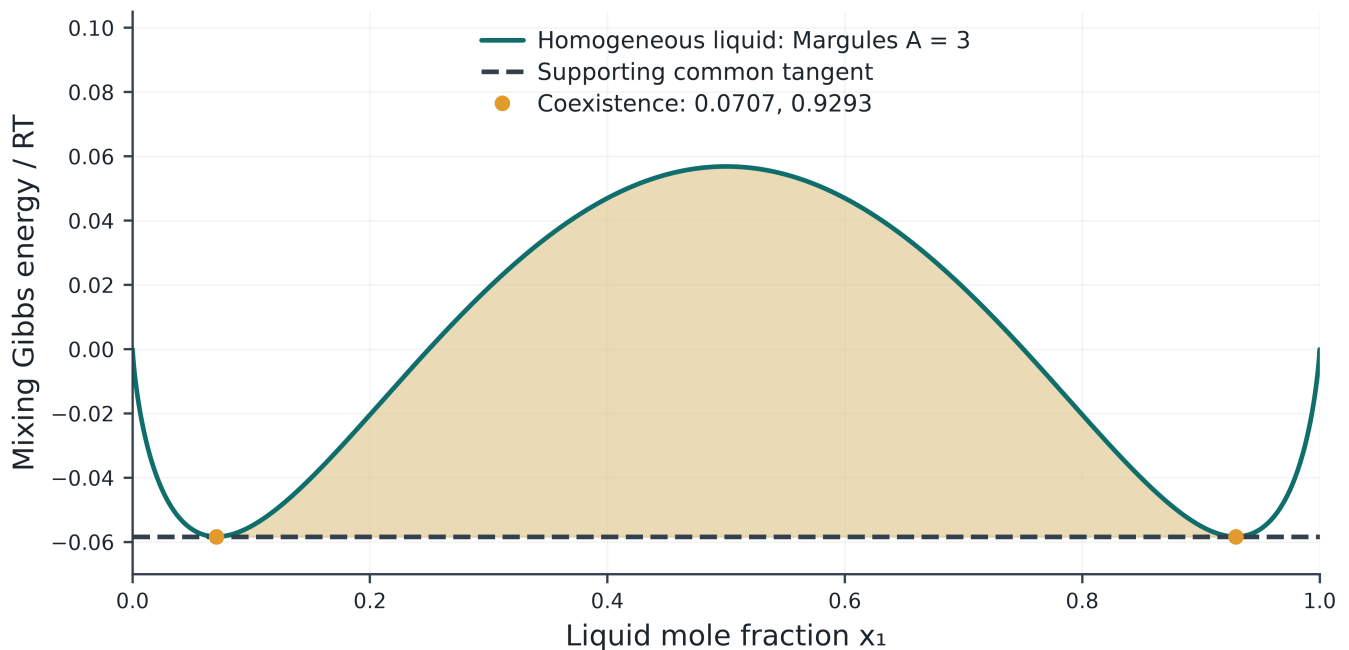
$$g'(x_\alpha) = g'(x_\beta) = [g(x_\beta) - g(x_\alpha)]/(x_\beta - x_\alpha)$$

$$f_\beta = (z_1 - x_\alpha)/(x_\beta - x_\alpha); f_\alpha = 1 - f_\beta$$

One example to reason through

Synthetic teaching values. These numbers illustrate the assumptions; they are not measurements of a real material.

For a given hypothetical supported tie line $x_\alpha=0.2$ and $x_\beta=0.8$, $z_1=0.5$ gives $f_\beta=0.5$. If z_1 changes to 0.35 at the same coexistence conditions, $f_\beta=0.25$ and $f_\alpha=0.75$. Check $0.75(0.2)+0.25(0.8)=0.35$. These endpoints are specified for arithmetic, not outputs of the model plotted below.



Check and interpret

The figure illustrates a different synthetic Margules model. Its tie-line endpoints come from the common tangent. Do not copy the numerical endpoints from the hand example into that model. State which evidence came from calculation and which values were supplied.

Bring three short answers

A sketch, a few numbers or one sentence is enough. Do this before opening the calculator; revise your prediction in class.

1. Why is a negative local curvature sufficient to reject local stability, while positive curvature alone does not prove global stability?

2. For the hand-example endpoints, predict what changes when z_1 moves from 0.5 to 0.35.

3. What would you inspect after a solver reports equal chemical potentials?

Optional extension

Partial molar properties provide a useful bridge: for binary molar $m(x)$, $\bar{m}_1 = m + (1-x)m'$ and $\bar{m}_2 = m - xm'$. Sketch a tangent and read its intercepts. Verify $x\bar{m}_1 + (1-x)\bar{m}_2 = m$. This is an optional paper exercise, not a new lab. LLE fitting and VLLE are further extensions.

Continue in the core lab

<https://www.skhgroup.net/teaching/2105603/labs/stability.html>

Reading locator: supplied textbook contents, sections 11.1-11.2, 14.11, 16.2-16.4.

Original course explanation and examples, not a reproduction of the textbook. Section numbers are based on the supplied contents pages. Required weekly independent work: 15–20 minutes preparation plus 30–40 minutes follow-up; optional extensions are additional.